

CLIMATE DATA PROCESSING AND CLIMOGRAPH CONSTRUCTION

CORON, RP (1991-2020)

PART I. Data Cleaning

TABLE 1: Cleaned Dataset for Coron, RP (1991–2020)

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A	B	C	D	E	F	G	H	I	J	K	L	
1	DATE	STATION	LATITUDE	LONGITUDE	ELEVATION	TEMP °C	DEWP °C	SLP HPA	WDSP MPS	MAX °C	MIN °C	PRCP MM
2	01/01/1991	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
3	02/01/1991	CORON, RP	12.216666	120.2	60	27.44	24.5	1012	2.52	32.11	25.61	0
4	03/01/1991	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
5	04/01/1991	CORON, RP	12.216666	120.2	60	26.61	23.67	1011.6	2.52	31.22	24.61	0
6	05/01/1991	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
7	06/01/1991	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
8	07/01/1991	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
9	08/01/1991	CORON, RP	12.216666	120.2	60	27.22	24.33	1010.5	2.21	33.72	22.72	0
10	09/01/1991	CORON, RP	12.216666	120.2	60	26.22	24.33	1011.3	2.01	32.89	23	8.89
11	10/01/1991	CORON, RP	12.216666	120.2	60	28.67	24.22	1011.1	2.01	38	24.39	0
12	11/01/1991	CORON, RP	12.216666	120.2	60	27.78	24.78	1012.1	2.01	31.11	23.89	0
13	12/01/1991	CORON, RP	12.216666	120.2	60	26	24.17	1012.7	1.59	32.5	23.72	0
14	13/01/1991	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
15	14/01/1991	CORON, RP	12.216666	120.2	60	26.78	23.61	1012.8	2.01	30.11	25.39	0
16	15/01/1991	CORON, RP	12.216666	120.2	60	26.56	24.61	1013.8	2.01	33.11	24.28	0
17	16/01/1991	CORON, RP	12.216666	120.2	60	26.44	23.72	1013.4	2.26	31.78	24.78	0
18	17/01/1991	CORON, RP	12.216666	120.2	60	26.17	23.39	1013.4	2.01	31.78	23.61	0
19	18/01/1991	CORON, RP	12.216666	120.2	60	26.06	23.06	1012.5	1.34	31	22.5	0
20	19/01/1991	CORON, RP	12.216666	120.2	60	27.61	24.11	1012.4	2.26	31.22	23.28	0
21	20/01/1991	CORON, RP	12.216666	120.2	60	27.06	24.06	1011.9	3.19	32.11	24.89	0
22	21/01/1991	CORON, RP	12.216666	120.2	60	26.44	23.67	1012.8	1.49	28.28	25	0
23	22/01/1991	CORON, RP	12.216666	120.2	60	26	23.22	1011.1	0.77	32.5	23.61	0
24	23/01/1991	CORON, RP	12.216666	120.2	60	27.22	22.39	1011.8	1.85	32.5	20	0
25	24/01/1991	CORON, RP	12.216666	120.2	60	27.06	22.5	1011.5	2.62	32	25.89	0

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A	B	C	D	E	F	G	H	I	J	K	L	
10879	12/10/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10880	13/10/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10881	14/10/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10882	15/10/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10883	16/10/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10884	17/10/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10885	18/10/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10886	19/10/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10887	20/10/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10888	21/10/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10889	22/10/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10890	23/10/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10891	24/10/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10892	25/10/2020	CORON, RP	12.216666	120.2	60	26.44	24.5	1006.4	2.26	29.22	25	96.012
10893	26/10/2020	CORON, RP	12.216666	120.2	60	25.22	23.78	1002.1	6.48	25.61	24.61	109.982
10894	27/10/2020	CORON, RP	12.216666	120.2	60	26.33	NA	1005.9	4.27	27.22	24.61	22.098
10895	28/10/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10896	29/10/2020	CORON, RP	12.216666	120.2	60	28.56	24.11	1010.2	2.01	32.78	24.61	0
10897	30/10/2020	CORON, RP	12.216666	120.2	60	28.11	24.94	1007.9	3.24	31.29	25	0
10898	31/10/2020	CORON, RP	12.216666	120.2	60	27.39	23.61	1006.8	1.49	32.27	25	0
10899	01/11/2020	CORON, RP	12.216666	120.2	60	26.89	24.5	1004.8	2.88	27.21	25	1.524
10900	02/11/2020	CORON, RP	12.216666	120.2	60	26.94	24.11	1008	2.01	30.21	25.22	7.874
10901	03/11/2020	CORON, RP	12.216666	120.2	60	28.06	24.61	1010	2.26	30.61	25.22	0
10902	04/11/2020	CORON, RP	12.216666	120.2	60	26.83	24.5	1009.6	2.01	28.78	25.78	24.13
10903	05/11/2020	CORON, RP	12.216666	120.2	60	26.61	24.33	1010.9	1.23	28.61	24.61	1.27

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Note: Through inspection, a significant amount of data was found to be missing, particularly in the years 2002-2004, 2013-2015, and 2016-2020. Missing values were labeled as “NA” to clearly visualize gaps during the 30-year period. Now, due to the identified gaps, statistical analysis on Part II used a subset of “complete years” (>75% data availability) to ensure climatological accuracy.

PART II. Summary Statistics

TABLE 2: Summary Climatological Statistics (75% Threshold)

STATISTICS	VALUES	REMARKS
MEAN ANNUAL TEMPERATURE	28.0 °C	Average of all valid daily records.
TOTAL ANNUAL RAINFALL	2,122.1 mm	Average of years with >75% data completeness.
WETTEST MONTH	July	461.1 mm (Average monthly rainfall).
DRIEST MONTH	March	11.0 mm (Average monthly rainfall).
ANNUAL TEMPERATURE RANGE	1.83 °C	Difference between Apr (29.1°C) and Sep (27.3°C).
STANDARD DEVIATION	159.8	Indicates a very high seasonal variability.

Note: These statistics represent the long-term averages calculated from years with reliable data completeness (1991, 1992, 1994, 1995, 1996, 1998, 2000, 2009, 2010, 2011, 2012).

PART III. Graph Construction

Note: The chart displays the full historical rainfall record available in Coron from 1991-2020. Gaps in the data line indicate periods of missing observations from the original dataset. Peak values correspond to the annual Southwest Monsoon season.

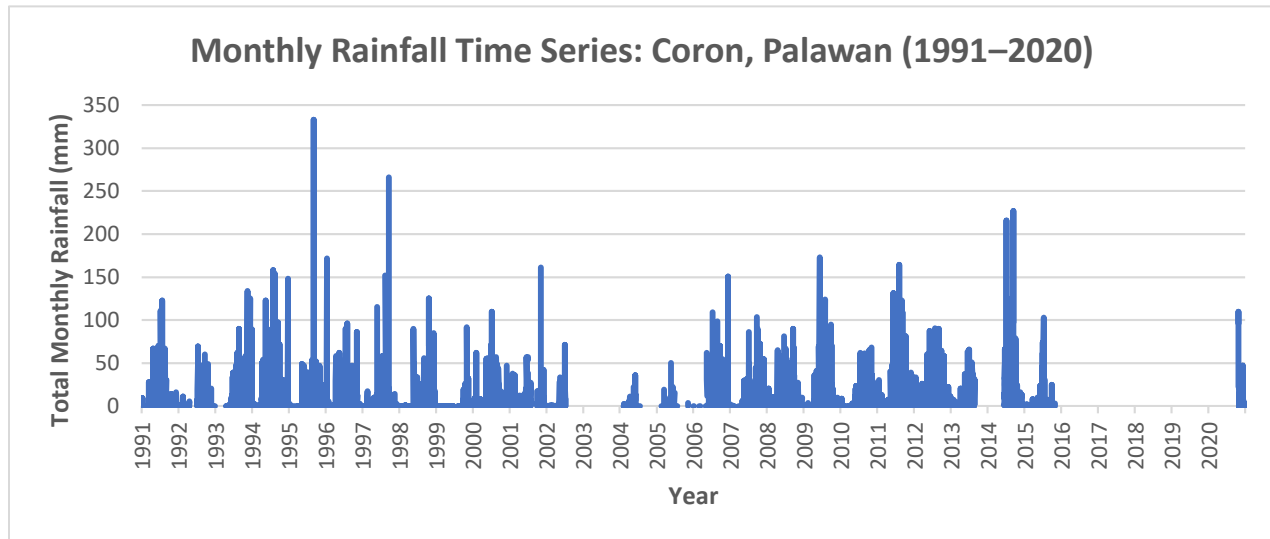


Figure 1: Monthly Rainfall Time Series (1991–2020)

Note: This figure illustrates the consistency of tropical temperatures in the region. The Y-axis has been adjusted (20°C–34°C) to visualize the subtle seasonal fluctuations. Data gaps match those identified on the first figure.

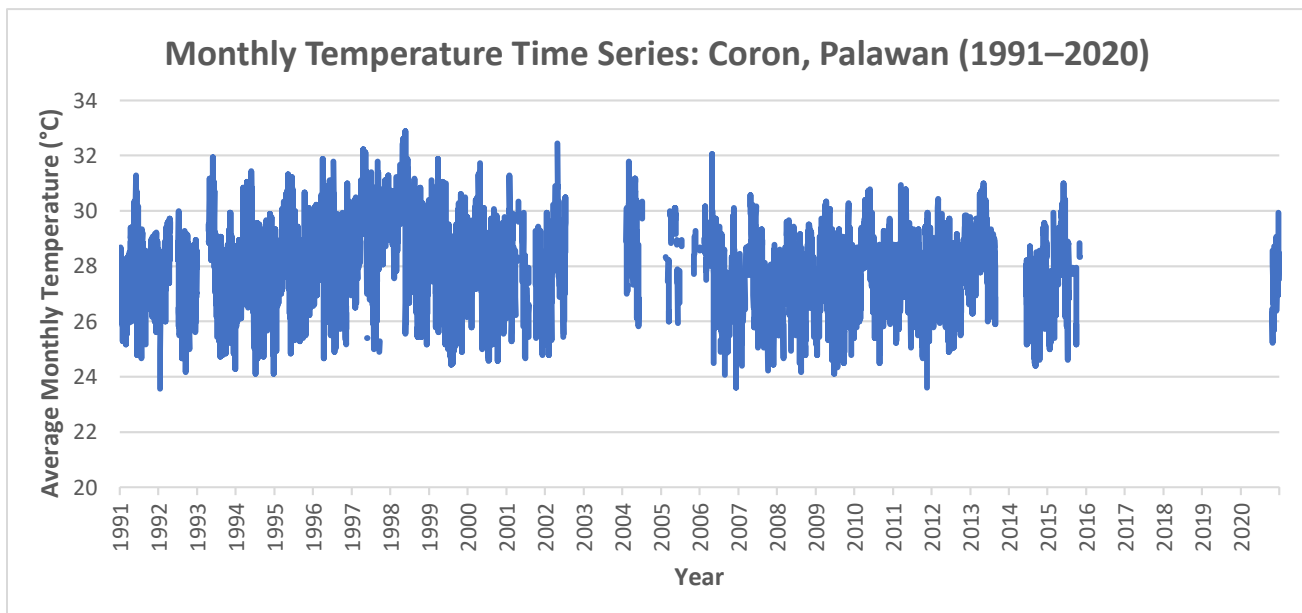


Figure 2: Monthly Temperature Time Series (1991–2020)

Note: This graph represents the "Climate Normal" for the station. Bars represent average monthly rainfall (left axis) while the line represents average monthly temperature (right axis). To ensure accuracy, this summary was calculated using only the years with $\geq 75\%$ data completeness

(1991, 1992, 1994, 1995, 1996, 1998, 2000, 2009, 2010, 2011, 2012).

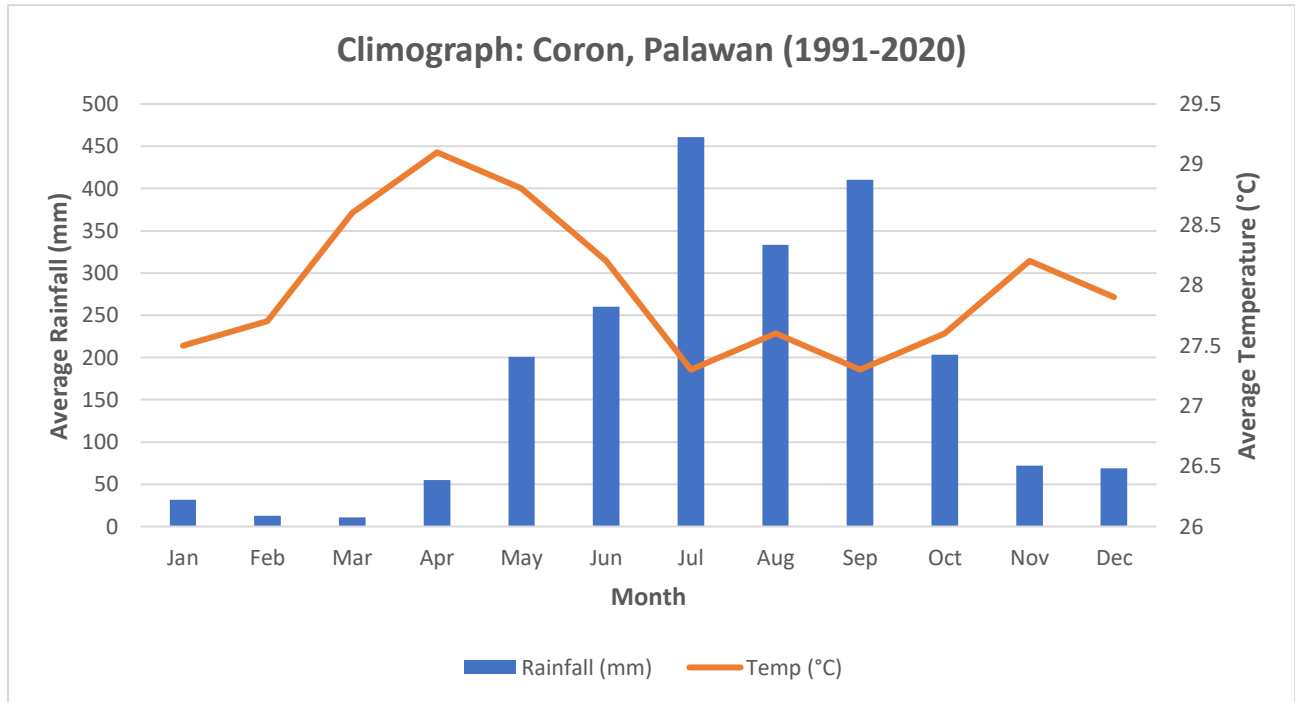


Figure 3: Climograph of Coron, Palawan (1991–2020)

PART IV. Interpretation

1. Describe the rainfall pattern of the station.
 - As shown in **Figure 3**, It has a distinct dry season from January to April and a heavy wet season from May to December which is driven by the Southwest Monsoon making it a Type I climate.
2. What months have peak rainfall?
 - Based on Table 2 and the bars in **Figure 3**, peak rainfall occurs in July with an average rainfall of around 461.1 mm and September with around 410.6 mm.
3. Is the climate monsoonal? Why?
 - Yes, because the extreme seasonal contrast seen in **Figure 1** such that *having a distinct wet and dry season*, and the high annual total rainfall in Table 2 (2,122.1 mm) suggests a monsoonal system.
4. Does temperature vary significantly throughout the year?
 - No. **Figure 2** shows that the temperature line stays within a narrow band. As noted in **Table 2**, the annual range is only 1.83°C, which is typical for a station in a tropical country.
5. Which variable shows greater variability: rainfall or temperature?
 - Rainfall shows significantly greater variability. As we can see from **Figure 3**, it visually contrasts the flat temperature line against the massive "peaks" of the rainfall bars, which fluctuate by over 400 mm annually.
6. If you were planning agriculture in this area, when would you plant crops? Why?
 - I should begin planting by late April or early May since it is the transition period from dry to wet season as seen in **Figure 3**. Through this, I can ensure that crops have consistent water during their critical growth phases.